

INTRO_TEXT

Dear Participant,

The objective of this expert survey is to identify preference elicitation and aggregation methods to select outcomes to measure the efficacy of medical treatments for depression.

This is a vignettes survey: with the help of your expertise, we will be able **to choose the most adequate preference elicitation method(s) and corresponding aggregation method(s)**. This expert survey does not test your scientific skills.

You can participate if you are **an academic or a researcher (with a PhD) with an interest in preference elicitation and/or aggregation methods.**

This survey consists of 4 parts, starting with demographics, followed by a section on preference elicitation methods, a section on aggregation methods, and ending with complementary questions. Responding to this survey **takes around 20 minutes**. This survey is

anonymous and no personal data will be collected.
Please take the time to complete each question truthfully.

As a counterpart, you have the opportunity to be cited in the acknowledgments of the future publication by sending us an email.

This study is a collaboration between Antonin Macé (PhD) and Jean-François Laslier (PhD) at the Paris School of Economics, Jérôme Lang (PhD) at Paris Dauphine University, and Dr. Astrid Chevance (MD, PhD) at the Center of Research in Epidemiology and Statistics, funded by the Fondation de France. Gregor Stolte, a student at the School of Business and Economics at the University of Maastricht is responsible for the survey. The protocol of the study is available on request and if you have any questions regarding the study, please contact us at gregor.stolte@student.maastrichtuniversity.nl

Question 1

Do you think you have expertise on preference elicitation and/or aggregation methods?

- ☐ Yes, on both preference elicitation and aggregation methods
- ☐ Yes, on preference elicitation methods
- ☐ Yes, on preference aggregation methods
- ☐ No
- ☐ I don't know

Question 2

Do you already have a PhD?

- ☐ Yes
- ☐ No
- ☐ I am still working on it

DEM_Q

Demographics

Question 1

I am a researcher working in the following field:

- ☐ Social choice theory
- ☐ Economics
- ☐ Sociology
- ☐ Political sciences
- ☐ Marketing
- ☐ Biomedical/Health Sciences
- ☐ Other

Question 2

What is your specific expertise within your research field?

Question 3

How many years have you worked in this field since your PhD?

- ☐ 0-5 Years
- ☐ 5-10 Years
- ☐ 10-15 Years
- ☐ >15 Years
- ☐ I do not wish to answer

Question 4

Regarding your expertise in methods to elicit individual preferences:

- ☐ I have designed at least one study of preference elicitation
- ☐ I have conducted in at least one study of preference elicitation
- ☐ I have developed at least one method to elicit preferences
- ☐ I have conducted theoretical/methodological studies on methods to elicit preferences
- ☐ None of the above

Question 5

Regarding your expertise in aggregation methods:

- ☐ I have designed at least one study of preference aggregation
- ☐ I have conducted at least one study of preference elicitation
- ☐ I have developed at least one method to aggregate individual preferences
- ☐ I have conducted theoretical/methodological studies on methods to aggregate individual preferences
- ☐ None of the above

Question 6

In which country do you currently work?

Question 7

What is your main affiliation?

- ☐ A University
- ☐ A Public research institute
- ☐ A Private research institute
- ☐ An Non-Governmental Organization (NGO)
- ☐ A Governmental agency
- ☐ An Inter-Governmental agency
- ☐ Industry/private organization
- ☐ Other : (please specify)

- ☐ I do not wish to answer

Question 8

How old are you?

- ☐ <30
- ☐ 30-45
- ☐ 46-60
- ☐ >60
- ☐ I do not wish to answer

Question 9

Gender Identity:

(if you do not wish to answer, please indicate **na** in the field below)

STUDY_BACKGROUND

In medical research, treatment effects are evaluated with “outcomes” such as symptoms (e.g., insomnia, anxiety, mood swings) and dimensions of functioning (e.g., autonomy, social isolation).

Our project focuses on **depression**, a disorder for which patients’ experiences and feelings about outcomes may be heterogeneous: some patients suffer more from insomnia, while others are mostly affected by anxiety, etc. In [previous research](#), we identified 80 potential outcomes for depression.

The goal of our project is to select 10 outcomes out of 80: this standardized set of outcomes should contain the 10 most important outcomes for depression, to be measured in **all** randomized controlled trials (of course specific trials might measure additional outcomes). Developing a standardized set of outcomes will allow the comparability and synthesis of research on

depression to **improve public health decisions and clinical recommendations.**

We plan to use a large-scale survey to elicit patient preferences over the 80 outcomes, and then aggregate these individual preferences to select the 10 outcomes.

Your expertise will be most valuable in helping us choose the most adequate preference elicitation method(s) and corresponding aggregation method(s), to select the 10 outcomes out of 80.

PREF_SECT

Elicitation Methods

This section of the expert survey deals with preference elicitation methods. We will present you with a range of preference elicitation methods and ask you to what extent you find them adequate for the ultimate task of selecting 10 outcomes for depression.

We identified **5 families of methods** to elicit preferences, reported in the table below. We would like to know your expert opinion on **the most adequate methods** to elicit patients preferences over the 80

outcomes.

In your deliberation, **please take in account these 2 constraints** regarding the elicitation method:

1. it should produce rich enough elicited preferences to allow for a relevant aggregation method (the final output is a set of 10 most representative outcomes out of 80).
2. it should minimally burden people with depression in terms of cognitive expectations and duration.

Question 1

Please rate each method according to the extent to which you believe it is adequate for the proposed task.

	Not at all	Rather not	Rather yes	Yes definitely	I don't know this method
A consensus method (e.g., Delphi survey)	☐	☐	☐	☐	☐
A choice-based method (e.g., discrete choice experiment, conjoint analysis , k-approval)	☐	☐	☐	☐	☐
A ranking method (e.g., full ranking, partial ranking, etc.)	☐	☐	☐	☐	☐
A rating method (scores , qualitative grades , etc.)	☐	☐	☐	☐	☐
A trade-off method (e.g. time trade-off , standard gamble, etc.)	☐	☐	☐	☐	☐

Question 2

If you would like to justify your choices from the table above, please do so in the text box below:

Question 3

On our side, we thought about a **two-step method** with 3 options to elicit patients' preferences:

- **Step A:** The patients are asked to select 15 outcomes out of 80 that are most important to them.
- **Step B – Option 1:** The patients are asked to measure the importance of the 15 outcomes they selected in step A, on a Likert scale (from 1 to 7: least important (1) to most important (7))
- **Step B – Option 2:** The patients are asked to distribute 150 points to the 15 outcomes they selected in step A, to measure their relative importance.
- **Step B – Option 3:** The patients are asked to rank the 15 outcomes into 5 levels in descending order of importance: 1 most important outcome, then 2 outcomes of second-order importance, then 3 outcomes at the 3rd level, etc.

What is your opinion on these proposed methods?

Feel free to suggest any limitations or improvements that may affect the adequacy of these methods to elicit patient preferences.

Question 4

To what extent would you recommend each of these methods for this task?

	Very Unlikely	Somewhat Unlikely	Somewhat Likely	Very Likely	N/A
Step A + Step B - Option 1	☐	☐	☐	☐	☐
Step A + Step B - Option 2	☐	☐	☐	☐	☐
Step A + Step B - Option 3	☐	☐	☐	☐	☐

Question 5

In light of all previous questions in this section, if you were in charge of designing the patient survey to select a subset of 10 outcomes out of 80, what preference elicitation method(s) would you favor? Please feel free

to suggest any other methods not mentioned in this expert survey.

AGGRE_SECT

Aggregation Methods

This section of the expert survey deals with preference aggregation methods. The objective here is to identify the most adequate method(s) to aggregate the individual preferences of patients over 80 outcomes. The ultimate goal is to select 10 outcomes, to be measured in all randomized controlled trials on depression.

We will present you with **4 families of methods** to aggregate individual preferences, reported in the table below. We would like to know your expert opinion on **the most adequate methods** to aggregate patients' preferences over the aforementioned 80 outcomes.

In your deliberation, **please take in account these 2 constraints** regarding the aggregation methods:

1. the method(s) should select the 10 outcomes deemed most important by patients
2. the 10 selected outcomes should be representative of the diversity of patient preferences

Question 1

Please rate each method according to the extent to which you believe it is adequate for the proposed task.

	Not at all	Rather not	Rather yes	Yes definitely	I don't know this method
A best-k method (e.g., Bloc Approval , Borda Rule , Range Voting)	☐	☐	☐	☐	☐
A proportional method (e.g., STV , PAV , Equal Shares , Phragmen)	☐	☐	☐	☐	☐
A method focused on diversity (e.g., Chamberlin- Courant)	☐	☐	☐	☐	☐
A majoritarian method (e.g., a Condorcet Comittee , near Condorcet Comittee)	☐	☐	☐	☐	☐

Question 2

If you would like to justify your choices from the table above, please do so in the text box below:

Question 3

On our side, we thought about an aggregation method called “**ordered-weighted average (OWA) with harmonic weights**” to select the subset of 10 outcomes. This method is versatile as it can be computed from various preference inputs. It is known to balance the objectives of efficiency and fairness, it satisfies in particular a property of **proportionality**, according to which each group of patients may obtain its fair share in the selected outcomes.

Formally, 10 outcomes are selected as follows: for each possible set S of 10 outcomes, we compute for each patient i , a valuation measuring the benefit for patient i that selecting exactly these outcomes would bring. This valuation is the sum of the importance attached by i to the outcomes in S , discounted by harmonic weights (1 for the most important outcome, $\frac{1}{2}$ for the second, $\frac{1}{3}$ for the third, etc.). We then select the set S^* that maximizes the aggregate valuation over all patients.

Consider for instance a simple example with 3 patients and 3 outcomes:

	o1	o2	o3
A	10	6	0
B	10	6	0
C	6	0	10

In this example, o1 is the outcome that is most important to patients A and B, and is also quite important to patient C. If 2 outcomes are to be selected, the OWA method will select o1o3, and not o1o2 (which maximizes total importance). The reason is that adding o3 to o1 is considered more valuable than adding o2, as the marginal benefit for patient C is more significant than that for patients A and B.

Aggregate valuation of o1o2: $2 * (10 + 6 / 2) + 6 = 32$

Aggregate valuation of o1o3: $2 * 10 + (10 + 6 / 2) = 33$

What is your opinion on the proposed aggregation method (OWA with harmonic weights)? Please feel free to elaborate on any limitations or ameliorations that could be implemented into the method.

Question 4

To what extent would you recommend this method for selecting the subset of 10 outcomes?

- ☐ Very Unlikely
- ☐ Somewhat Unlikely
- ☐ Somewhat Likely
- ☐ Very Likely

☐

We grouped the 80 outcomes that were meaningful for depressed patients into 9 categories: cognitive symptoms, perception of self, etc.. We could account for this categorization by adapting the aforementioned aggregation method in two steps.

In the first step, we would select the categories to be represented in the subset of 10 outcomes (for instance, requiring that the subset selects 2 outcomes from category 1 and 1 from category 3, etc.). In the second step, we would select the outcomes within each category. For both steps, the OWA method with harmonic weights presented earlier could be used.

Question 5

What do you think of this two-step aggregation method with respect to the method that does not rely on categories?

Question 6

In light of all previous questions in this section, if you were in charge of aggregating the preferences from the patient survey, what method(s) would you favor? Please feel free to suggest any other methods not mentioned in this expert survey.

COMP_QUESTION

Complementary Questions

There are 5 questions left in this expert survey. Please take the time to answer each question truthfully.

Question 1

Would you modify the proposed elicitation and/or aggregation methods to develop a standardized set of outcomes if the objective was to obtain a set of 25 outcomes out of 80, instead of 10 out of 80? If so, how?

Question 2

The development of standardized sets of outcomes has historically used the Delphi method. This method consists of several rounds where participants evaluate the importance of all outcomes on a Likert scale, and then receive feedback about how others evaluated the outcomes. Finally, participants are invited to revise their evaluations in subsequent rounds until a consensus is reached, and guidelines on consensually important outcomes can be established.

Would you recommend this method?

☐ Why?

☐ Why not?

Question 3

In the proposed elicitation method, we implicitly assumed that preferences over outcomes were separable. In practice, it could be that two outcomes

are particularly important when they simultaneously affect the same patient (thus, attenuating one outcome makes another outcome easier to endure). If we aimed to capture such interdependencies within our elicitation method, how would you proceed?



Question 4

Could you please help us in identifying further experts who might have valuable input for our expert survey? You may select up to 5 boxes in this question (one for each suggested expert). This information will be used to contact new respondents.

- ☐ Expert 1
- ☐ Expert 2
- ☐ Expert 3
- ☐ Expert 4
- ☐ Expert 5
- ☐ I do not wish to answer

Question 5

Please let us know if you have any final comments for us. Thank you very much for your participation!

If you would like to be mentioned in the

acknowledgements of this expert survey, please send
us an email at
gregor.stolte@student.maastrichtuniversity.nl

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